Chapter 1. Probability Distributions

Axioms, conditional probability, random variables, moments, generating functions, inequalities

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1.1 Probability and its properties

A **sample space** S is the set of all outcomes of an experiment; **events** are subsets of S. A probability P assigns numbers to events subject to Kolmogorov's axioms:

(A1) $P(A) \geq 0$ for every event A;

(A2) $P(S) = 1$;

(A3) (countable additivity) if $A_1, A_2, \ldots$ are pairwise disjoint, $P(\bigcup_i A_i) = \sum_i P(A_i)$.

Theorem 1.1

(a) $P(\emptyset) = 0$, $0 \leq P(A) \leq 1$, and $P(A^c) = 1 - P(A)$.

(b) (Monotonicity) $A \subseteq B \Rightarrow P(A) \leq P(B)$.

(c) (Inclusion–exclusion) $P(\bigcup_{i=1}^n A_i) = \sum_i P(A_i) - \sum_{i<j} P(A_i \cap A_j) + \cdots + (-1)^{n-1} P(A_1 \cap \cdots \cap A_n)$.

(d) (Countable subadditivity) $P(\bigcup_i A_i) \leq \sum_i P(A_i)$.

Proof. (a) Apply (A3) to $S = S \cup \emptyset \cup \emptyset \cup \cdots$ to get $P(\emptyset) = 0$, then to the disjoint union $S = A \cup A^c$. (b) $B = A \cup (B \setminus A)$ disjointly. (c) For $n = 2$, $A_1 \cup A_2 = (A_1 \setminus A_2) \cup (A_2 \setminus A_1) \cup (A_1 \cap A_2)$ and $A_k = (A_k \setminus A_j) \cup (A_1 \cap A_2)$; combine. The general case follows by induction on n. (d) Let $B_1 = A_1$, $B_n = A_n \setminus (A_1 \cup \cdots \cup A_{n-1})$. The B_n are disjoint with the same union and $B_n \subseteq A_n$, so $P(\bigcup A_i) = \sum P(B_n) \leq \sum P(A_n)$. ∎

Theorem 1.2 (Continuity of probability)

If $A_1 \subseteq A_2 \subseteq \cdots$ then $P(\bigcup_n A_n) = \lim_n P(A_n)$; if $B_1 \supseteq B_2 \supseteq \cdots$ then $P(\bigcap_n B_n) = \lim_n P(B_n)$.

Proof. With $C_1 = A_1$, $C_n = A_n \setminus A_{n-1}$ (disjoint, same union),

$$P\Big(\bigcup_n A_n\Big) = \sum_k P(C_k) = \lim_n \sum_{k \leq n} P(C_k) = \lim_n P(A_n).$$

. For decreasing B_n, apply this to $A_n = B_n^c$ and take complements. ∎

For example, under the uniform probability $P((a, b]) = b - a$ on $[0, 1]$, $P(\{b\}) = \lim_n P((b - \frac{1}{n}, b]) = 0$: single points have probability zero.

1.2 Conditional probability and independence

Definition (Conditional probability)

For $P(A) > 0$, $P(B \mid A) = P(A \cap B)/P(A)$. As a function of B it is again a probability (on the reduced sample space A).

Theorem 1.3

(a) (Multiplication) $P(A \cap B) = P(B \mid A) P(A)$.

(b) (Total probability) If $A_1, A_2, \ldots$ partition S with $P(A_i) > 0$, then $P(B) = \sum_i P(B \mid A_i) P(A_i)$.

(c) (Bayes) In the same setting, with $P(B) > 0$,

$$P(A_j \mid B) = \frac{P(B \mid A_j)P(A_j)}{\sum_i P(B \mid A_i)P(A_i)} \;\propto\; P(B \mid A_j)P(A_j).$$

Proof. (a) is the definition. (b) $B = \bigcup_i (B \cap A_i)$ disjointly; use (A3) and (a). (c) $P(A_j \mid B) = P(B \cap A_j)/P(B)$; expand the numerator by (a) and the denominator by (b). ∎

$P(A_j)$ are the **prior** and $P(A_j \mid B)$ the **posterior** probabilities; Bayes' theorem converts the first into the second after observing B.

Example 1.1

A disease has prevalence 1%. A test detects it with probability 0.95 and gives a false positive with probability 0.05. Then

$$P(\text{disease} \mid +) = \frac{0.95 \cdot 0.01}{0.95 \cdot 0.01 + 0.05 \cdot 0.99} = \frac{0.0095}{0.059} \approx 0.16.$$

A positive result raises the probability from 1% only to about 16%, because healthy people vastly outnumber sick ones.

Definition (Independence)

A, B are independent if $P(A \cap B) = P(A)P(B)$. Events $A_1, \dots, A_n$ are (mutually) independent if $P(A_{i_1} \cap \cdots \cap A_{i_k}) = P(A_{i_1}) \cdots P(A_{i_k})$ for *every* subcollection.

If A, B are independent then so are A, B^c: $P(A \cap B^c) = P(A) - P(A)P(B) = P(A)P(B^c)$. Pairwise independence does not imply mutual independence.

Example 1.2

Toss two fair coins: $A = \{\text{first is H}\}$, $B = \{\text{second is H}\}$, $C = \{\text{the two agree}\}$. Each has probability $\frac{1}{2}$ and each pair intersects with probability $\frac{1}{4}$, so the events are pairwise independent; but $P(A \cap B \cap C) = \frac{1}{4} \neq \frac{1}{8}$.

1.3 Random variables and distributions

A **random variable** X is a real-valued function on S; events such as $\{X \le x\}$ inherit probabilities.

- **Discrete** X takes values $x_1, x_2, \dots$; its probability mass function (pmf) $f(x) = P(X = x)$ satisfies $f \ge 0$, $\sum_x f(x) = 1$, $P(X \in A) = \sum_{x \in A} f(x)$.
- **Continuous** X has a probability density function (pdf) $f \ge 0$ with $\int f = 1$ and $P(a \le X \le b) = \int_a^b f(x)\,dx$. Then $P(X = a) = 0$, and $P(a \le X \le a + \Delta) \approx f(a)\Delta$; $f(a)$ itself is not a probability.

We write $X \sim f$. The indicator $I_A(x) = I(x \in A)$ gives compact formulas such as $f(x) = 2x\, I_{[0,1]}(x)$.

1.4 Expectation, variance and moments

Definition (Expectation)

$E[g(X)] = \sum_x g(x)f(x)$ or $\int g(x)f(x)\,dx$, defined when the sum or integral converges absolutely. The mean is $\mu = E(X)$, the variance $\operatorname{Var}(X) = E[(X-\mu)^2]$, the standard deviation $\operatorname{sd}(X) = \sqrt{\operatorname{Var} X}$.

Theorem 1.4

(a) $E(aX+b) = aE(X)+b$ and $E[c_1g_1(X)+c_2g_2(X)] = c_1Eg_1(X)+c_2Eg_2(X)$. (b) If $g_1 \le g_2$ then $Eg_1(X) \le Eg_2(X)$. (c) $\operatorname{Var}(aX+b) = a^2 \operatorname{Var} X$ and $\operatorname{Var} X = E(X^2) - (EX)^2$.

Proof. (a), (b) are linearity and monotonicity of sums and integrals. (c) $\operatorname{Var}(aX+b) = E[a^2(X-\mu)^2]$; and $E(X-\mu)^2 = E(X^2) - 2\mu EX + \mu^2 = E(X^2) - \mu^2$. ■

The **standardized** variable $Z = (X-\mu)/\sigma$ has $EZ = 0$, $\operatorname{Var} Z = 1$. Means need not exist: for $f(k) = \frac{1}{k(k+1)}$, $k = 1, 2, \ldots$, $\sum kf(k) = \sum \frac{1}{k+1} = \infty$.

Example 1.3

For $f(x) = 2x$ on $[0,1]$: $EX = \int_0^1 2x^2\,dx = \frac{2}{3}$, $EX^2 = \int_0^1 2x^3\,dx = \frac{1}{2}$, so $\operatorname{Var} X = \frac{1}{2} - \frac{4}{9} = \frac{1}{18}$.

1.5 Distribution functions and generating functions

The **cumulative distribution function** (cdf) is $F(x) = P(X \le x)$. For a discrete X it is a step function with jumps $f(x_k)$ at the x_k; for a continuous X, $F(x) = \int_{-\infty}^x f$ and $F' = f$ at continuity points of f.

Theorem 1.5

(a) F is nondecreasing; (b) $F(-\infty) = 0$, $F(+\infty) = 1$; (c) F is right-continuous; (d) $F(a-) = P(X < a)$, so $F(a) - F(a-) = P(X = a)$.

Proof. (a) $\{X \le x_1\} \subseteq \{X \le x_2\}$. (b), (c), (d) follow from Theorem 1.2 with $\{X \le -k\} \downarrow \emptyset$, $\{X \le k\} \uparrow S$, $\{X \le x + \frac{1}{n}\} \downarrow \{X \le x\}$, $\{X \le a - \frac{1}{n}\} \uparrow \{X < a\}$. ■

A cdf can mix both types, e.g. a waiting time that equals a constant with probability $\frac{1}{2}$ and is uniform otherwise has a jump plus a continuous part.

Probability generating function. For X with values in $\{0, 1, 2, \ldots\}$, $G(s) = E(s^X) = \sum_x s^x f(x)$, $|s| < 1$. It determines the distribution: $f(k) = G^{(k)}(0)/k!$.

Moment generating function. If $E(e^{tX}) < \infty$ for all t in an open interval $(-h, h)$, the mgf is $M(t) = E(e^{tX})$. The kth moment is $m_k = E(X^k)$.

Theorem 1.6

If M exists on $(-h, h)$, then (a) all moments exist, $E(X^k) = M^{(k)}(0)$ and $M(t) = \sum_k \frac{E(X^k)}{k!} t^k$ near 0; (b) if $M_X = M_Y$ on a neighborhood of 0, then X and Y have the same distribution.

Proof of (a). Fix $|t| < h$. Since $|tx|^k/k! \le e^{|tx|} \le e^{tx}+e^{-tx}$, we get $|t|^k E|X|^k/k! \le M(t)+M(-t) < \infty$, so all moments exist. Moreover the partial sums of $\sum_k (tX)^k/k!$ are dominated by $e^{|tX|}$, which is integrable. By dominated convergence $M(t) = E\sum_k \frac{(tX)^k}{k!} = \sum_k \frac{E(X^k)}{k!}t^k$. A power series may be differentiated term by term inside its interval of convergence, so $M^{(k)}(0) = E(X^k)$. Part (b) (uniqueness) is proved with characteristic functions and is beyond these notes. ■

All moments can exist without an mgf (the lognormal distribution).

Example 1.4

For the geometric pmf $f(k) = p(1-p)^{k-1}$, $k = 1, 2, \ldots$: $M(t) = \sum_k e^{tk}p(1-p)^{k-1} = \dfrac{pe^t}{1-(1-p)e^t}$ for $(1-p)e^t < 1$. Then $M'(0) = 1/p$ is the mean.

Cumulants. $C(t) = \log M(t) = \sum_{r\ge 1} c_r \frac{t^r}{r!}$ defines the cumulants c_r. Expanding $\log(1+A) = A - A^2/2 + \ldots$ with $A = m_1 t + m_2 t^2/2 + \ldots$ gives $c_1 = m_1 = \mu$ and $c_2 = m_2 - m_1^2 = \sigma^2$. For the standard exponential, $M(t) = (1-t)^{-1}$, $C(t) = -\log(1-t)$, and $c_r = (r-1)!$.

1.6 Inequalities

Theorem 1.7 (Jensen)

If ϕ is convex on an interval I containing the values of X and EX exists, then $\phi(EX) \le E\phi(X)$.

Proof. A convex function lies above a supporting line at $\mu = EX$: $\phi(x) \ge \phi(\mu) + c(x-\mu)$ for some c and all $x \in I$ (for twice differentiable ϕ, take $c = \phi'(\mu)$ and use Taylor with $\phi'' \ge 0$). Take expectations. ■

E.g. $(EX)^2 \le E(X^2)$, and $E\log X \le \log EX$ for $X > 0$.

Theorem 1.8 (Lyapunov)

If $E|X|^s < \infty$ and $0 < r < s$, then $(E|X|^r)^{1/r} \le (E|X|^s)^{1/s}$.

Proof. $\phi(y) = y^{s/r}$ is convex on $[0,\infty)$. With $Y = |X|^r$ (integrable since $|X|^r \le 1 + |X|^s$), Jensen gives $(EY)^{s/r} \le E(Y^{s/r}) = E|X|^s$. ■

Theorem 1.9 (Markov and Chebyshev)

(a) If $E|Z|^r < \infty$ $(r > 0)$, then $P(|Z| \ge k) \le E|Z|^r/k^r$ for $k > 0$. (b) If $\operatorname{Var} X < \infty$, then $P(|X - EX| \ge k) \le \operatorname{Var} X/k^2$.

Proof. (a) $I(|Z| \ge k) \le (|Z|/k)^r$ pointwise; take expectations. (b) Apply (a) to $Z = X - EX$ with $r = 2$. ■

So $P(|X-\mu| \ge 3\sigma) \le \frac{1}{9}$ for *every* distribution with finite variance (for the normal the exact value is 0.0027).

Theorem 1.10

If $\operatorname{Var} X = 0$ then $P(X = \mu) = 1$.

Proof. $\{|X-\mu|>0\} = \bigcup_n \{|X-\mu| \geq \frac{1}{n}\}$, and each set has probability $\leq \operatorname{Var} X \cdot n^2 = 0$ by Chebyshev; countable subadditivity finishes. ■